SecureCorp Security Assessment Engagement

Research Log

Prepared by: Junior Security Consultant (Student)

Note on this log: this record was reconstructed after the fact from what each weekly report already documented as a resolved blocker. For several entries, the specific research steps taken at the time (what was searched, which source helped) were not recorded as they happened, and are marked below as "Not recorded at the time" rather than invented. This is an honest gap in note-taking discipline during the engagement, not a gap in the troubleshooting itself — each issue below was in fact identified and resolved, with the fix confirmed in the corresponding weekly report.

Log

Week / Task	What Broke	What Fixed It	Source / Method	Notes
Week 1, Task 1.1	Target VM would not boot — UEFI/BIOS boot mode mismatch	Corrected the VM's boot firmware setting to match the OVA's expected mode	Not recorded at the time	Resolved; documented only as a resolved blocker in the Week 1 progress note
Week 1, Task 1.5	rsyslog on the log box was not receiving forwarded logs — AppArmor and directory-ownership issues	Adjusted AppArmor profile / directory permissions so rsyslog could write to the log path	Not recorded at the time	Resolved; documented only as a resolved blocker in the Week 1 progress note
Week 1, Task 1.5	Log forwarding config on the target had a syntax error, so logs never left the target	Corrected the syntax in the rsyslog forwarding config file	Not recorded at the time	Resolved; documented only as a resolved blocker in the Week 1 progress note
Week 2, Task 2.8	Hydra could not complete the SSH handshake against the target — key-exchange / MAC algorithm mismatch with the legacy OpenSSH version	Added KexAlgorithms, HostKeyAlgorithms, and MACs overrides to the attacker's SSH client config to support the legacy algorithms	Not recorded at the time	Confirmed fix — full error output and working config saved in the Week 2 evidence screenshot
Week 1 / Week 2 boundary	All three lab VMs (Kali, target, log box) were running on UTM's Shared Network mode instead of a properly isolated network, which doesn't fully meet the engagement's isolation requirement	Switched all three VMs' Network Mode from Shared Network to Host Only (Default private); re-verified all static IPs and inter-VM connectivity afterward with no reassignment needed	Identified and corrected collaboratively while reviewing the lab for this report; UTM's own Network settings panel was used to diagnose and change the mode	Confirmed fix — documented with before/after ping evidence in Week 1, "Network Mode Correction"

Reflection on Research Practice

Going forward, the habit this log makes clear is worth keeping: when a blocker is hit and resolved, write down the error, what was tried, and what worked, at the time — not after. Several of the fixes above were real and effective, but because the research trail wasn't captured in the moment, this log can only state the outcome, not the method, which is a weaker form of evidence than the handbook's own research-log guidance (Section 6) asks for.